GEOMETRIC NORMALITY OF THE NORMALIZATION IS REACHED AFTER A FINITE PURELY INSEPARABLE FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The last sentence of the proof is the author's own 'It follows readily from this that nu is the normalization morphism'; the sentence is reproduced verbatim, and the two properties it refers to were both established inside the counted span.

Editorial note. The counted spans carry 7 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-finite-extension-geometrically-normal is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Normalization of a scheme locally of finite type over a field. Same chapter, Lemma lemma-normalization-locally-algebraic, invoked at proof lines 5001-5002. Supplies the normalization morphism over the perfect closure.
- Descent of finite presentation, finiteness, open subschemes and closed immersions along a filtered colimit. Limits, Lemmas limits-lemma-descend-finite-presentation, limits-lemma-descend-finite-finite-presentation, limits-lemma-descend-opens and limits-lemma-descend-closed-immersion-finite-presentation, invoked at proof lines 5003-5005, 5009-5010 and 5018-5021. Supplies each of the objects and properties descended to a finite subextension.
- Geometric normality is preserved when passing to a larger base field. Same chapter, Lemma lemma-geometrically-normal, invoked at proof lines 5007-5008. Supplies that the descended morphism is geometrically normal over k'.
- A birational morphism is an isomorphism over a dense open. Morphisms, Lemma morphisms-lemma-birational-isomorphism-over-dense-open, invoked at proof lines 5015-5017. Supplies the dense open on which the normalization is a closed immersion.

Source reference. Original lemma 038O: sources/stacks-varieties/pinned-original-source.tex, source lines 2160–2160. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0BXR: sources/stacks-varieties/pinned-original-source.tex, source lines 4903–4903. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01ZO: sources/stacks-chow/tags.txt, source lines 2601–2601. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01Z4: sources/stacks-chow/tags.txt, source lines 2579–2579. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BAJ: sources/stacks-chow/tags.txt, source lines 14212–14212. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be an algebraic k-scheme. Then there exists a finite purely inseparable extension k'/k such that the normalization Y of $X_{k'}$ is geometrically normal over k'.*

Proof. Let $K = k^{perf}$ be the perfect closure. Let Y_K be the normalization of X_K, see Lemma 0BXR. By Limits, Lemma 01ZM there exists a finite sub extension $K/k'/k$ and a morphism $\nu : Y \to X_{k'}$ of finite presentation whose base change to K is the normalization morphism $\nu_K : Y_K \to X_K$. Observe that Y is geometrically normal over k' (Lemma 038O). After increasing k' we may assume $Y \to X_{k'}$ is finite (Limits, Lemma 01ZO). Since $\nu_K : Y_K \to X_K$ is the normalization morphism, it induces a birational morphism $Y_K \to (X_K)_{red}$. Hence there is a dense open $V_K \subset X_K$ such that $\nu_K^{-1}(V_K) \to V_K$ is a closed immersion (inducing an isomorphism of $\nu_K^{-1}(V_K)$ with $V_{K,red}$, see for example Morphisms, Lemma 0BAJ). After increasing k' we find V_K is the base change of a dense open $V \subset Y$ and the morphism $\nu^{-1}(V) \to V$ is a closed immersion (Limits, Lemmas 01Z4 and 01ZP). It follows readily from this that ν is the normalization morphism and the proof is complete. □